

Active Learning–Guided Adaptive ESR Acquisition for Rapid Quantum Spin Characterisation

Srishti Tripathi, Siddhant Singh, Akhil Nirala, Dr. Aruna Shukla

Dr. Shyama Prasad Mukherjee International Institute of Information Technology, Naya Raipur, India

*Presenter email address: srishti24100@iiitnr.edu.in

Abstract

Electron Spin Resonance (ESR) is a quantum spectroscopy technique for investigating unpaired electron spins, which form the basis of two-level quantum systems. In practical studies of quantum materials and spin-based platforms, acquiring high-quality ESR spectra can be time-consuming because dense field sweeps and repeated signal averaging are often required, especially in low signal-to-noise regimes. This reduces experimental throughput and slows rapid spin characterisation.

Conventional ESR workflows rely on fixed, uniform scanning followed by spectral fitting and post-processing. While widely used, these approaches often spend substantial time sampling low-information regions and may become unreliable when spectra are weak or noisy. In this work, we propose an active learning–guided adaptive ESR acquisition framework that treats ESR as an information-driven measurement process and selects the next most informative measurements instead of scanning the full spectrum uniformly.

Physics-informed ESR simulations are used to generate realistic spectra across varying spin parameters and conditions, and a deep neural network is trained to infer spin signatures from partial measurements. An acquisition policy iteratively chooses field regions or sampling points that maximize information gain, enabling a closed-loop measure–infer–select strategy. The proposed approach reduces the required scans while preserving key spin features such as resonance position and linewidth across different noise regimes. By shifting ESR from fixed acquisition to adaptive measurement planning, this method supports rapid ESR-based quantum sensing and provides a scalable software pathway toward intelligent, measurement-aware ESR experimentation.

Keywords: Electron Spin Resonance, Active Learning-Based Acquisition, Adaptive Spectral Sampling, Quantum Spin Characterisation, Intelligent ESR Instrumentation

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